Week 13

Machine Learning and Big Data - DATA622

Fall 2022

CUNY School of Professional Studies

Week 13

1. Discussion Board Week 13: Neural Network
2. Reading materials:
   - **Textbook Reading: Practical Machine Learning in R (PMLiR)**
   - **ML Concepts Reading & Videos**
     - Read: Reconciling Modern Machine Learning Practice and the classical Bias-Variance trade-off
     - Read: A farewell to the bias-variance trade-off?

No free lunch Theorem

Bias-free learning is futile

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TANSTAAFL

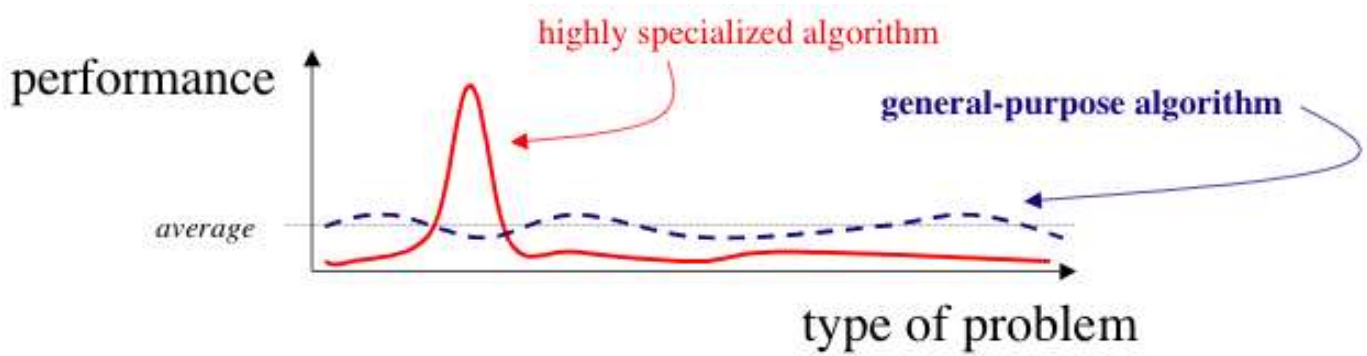
- No-Free-Lunch Theorem states:
  - No single classifier works the best for all possible problems
  - We need to make assumptions to generalize (we need bias)

TANSTAAFL

- Theorem:

  The average performance of any pair of algorithms across all possible problems is identical.

- If an algorithm achieves superior results on some problems, then it must pay with inferiority on the other problems



Source: https://medium.com/@LeonFedden/the-no-free-lunch-theorem-62ae2c3ed10c

Bias

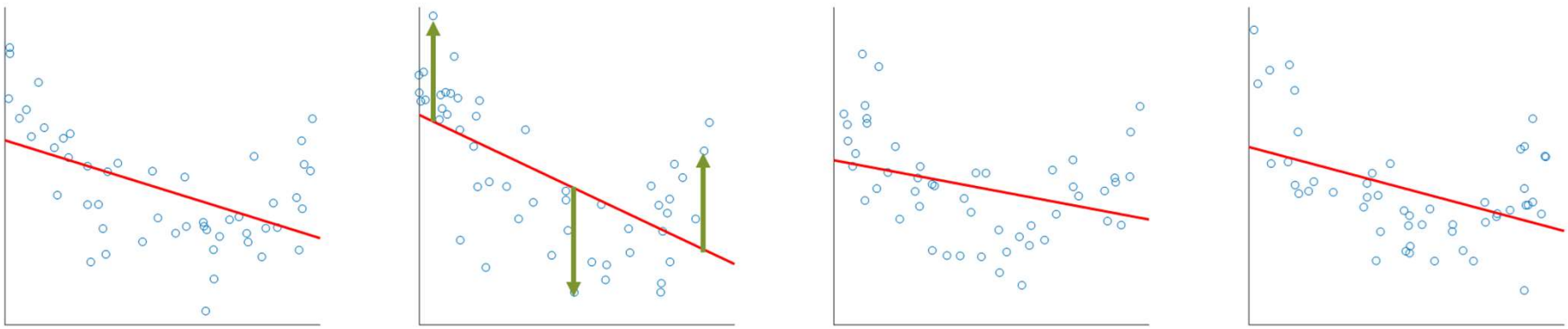
Complexity of models

Bias

- Dictionary definition:
  - Bias: Prejudice in favor of or against one thing
  - Prejudice: Preconceived opinion
- Statistics definition:
  - Bias: The tendency of a statistic to overestimate or underestimate the population parameter you're trying to measure
  - Bias: The approach that leads to a systematic difference between the true parameters of a population and the statistics used to estimate those parameters
- Machine Learning definition:
  - Bias is the difference between the average prediction of our model and the correct value which we are trying to predict.
  - Model with high bias pays very little attention to the training data and oversimplifies the model. It always leads to high error on training data.

Bias

- Regardless of training sample, or size of training sample, model will produce consistent errors



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How is bias introduced?

At each step of the ML Lifecycle you must make decisions which will impact the

- Data Acquisition
  Data is collected from various sources like files, databases, etc. and merged into one medium. What datasets you select will impact your model.
- Data Exploration
  Understanding and exploring data in order to identify the .
- Data Preparation
  To use data for training, some data processing needs to be done. This includes data cleaning for duplicates, processing missing data, removing noisy data, etc.
- Feature Engineering
  The goal of feature engineering is to convert categorical and ordinal data into numerical features. Includes feature creation, and overlaps with data preparation.
- Model Selection
  Selecting the correct model. Research needs to be done to finalize which model will work best for the dataset. At this step, the model is trained, interpreted, and evaluated for best performance. Key impact to bias.
- Model Training
  At this step, the model is trained, interpreted, and evaluated for best performance. Key impact to bias.
- Hyperparameter Tuning
  (Optional) Hyperparameter tuning to improve the performance of models by fine-tuning the parameters electing the correct model.
- Prediction (Serving)
  Make predictions of the unseen data. How you serve the model, monitor the model, and the decay of the model will impact its accuracy.

Variance

Generalizing the model

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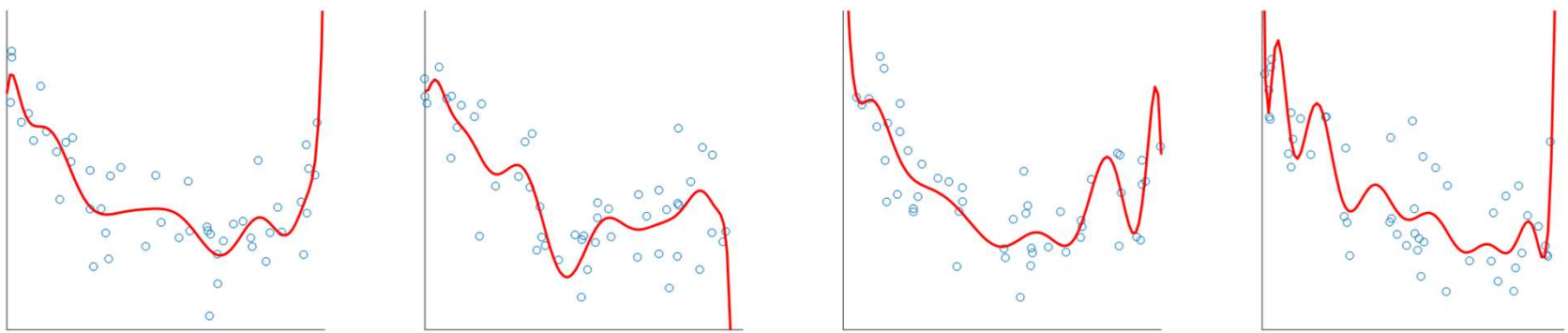
Variance

- Inability to perfectly estimate parameters from limited data
- Dictionary definition:
  - Bias: Prejudice in favor of or against one thing
  - Prejudice: Preconceived opinion
- Statistics definition:
  - Bias: The tendency of a statistic to overestimate or underestimate the population parameter you're trying to measure
  - Bias: The approach that leads to a systematic difference between the true parameters of a population and the statistics used to estimate those parameters
- Societal bias is different to machine learning/statistical bias
- Bias is not intrinsically bad - if it is suitable for the problem domain
- Societal bias must be minimized. Machine Learning cannot exist without bias.

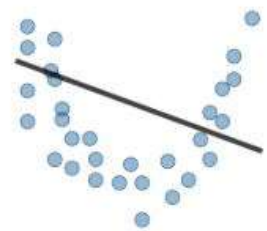
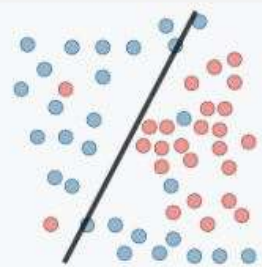
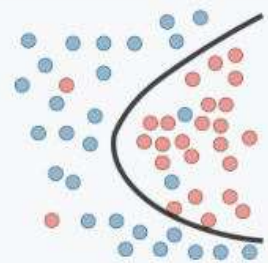
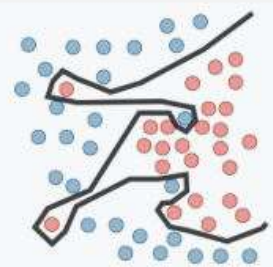
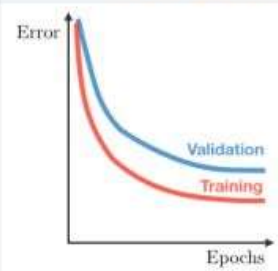
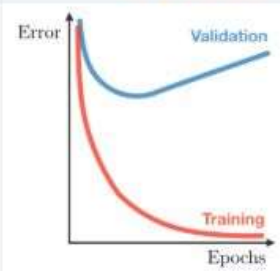
Variance

- Different samples of training data yield different model fits



Over & under-fitting

- Reducing Underfitting
  1. Increase model complexity
  2. Increase number of features
  3. Remove noise from the data
  4. Increase the number of epochs / increase the duration of training
- Reducing overfitting
  1. Increase training data.
  2. Reduce model complexity
  3. Early stopping during the training phase
  4. Adding noise to the data
  5. Regularization
  6. Use dropout for neural networks to tackle overfitting.

	Underfitting	Just right	Overfitting
Symptoms	• High training error • Training error close to test error • High bias	• Training error slightly lower than test error	• Very low training error • Training error much lower than test error • High variance
Regression illustration			
Classification illustration			
Deep learning illustration			
Possible remedies	• Complexify model • Add more features • Train longer		• Perform regularization • Get more data

Bias-Variance Trade-off

Bias-free learning is futile

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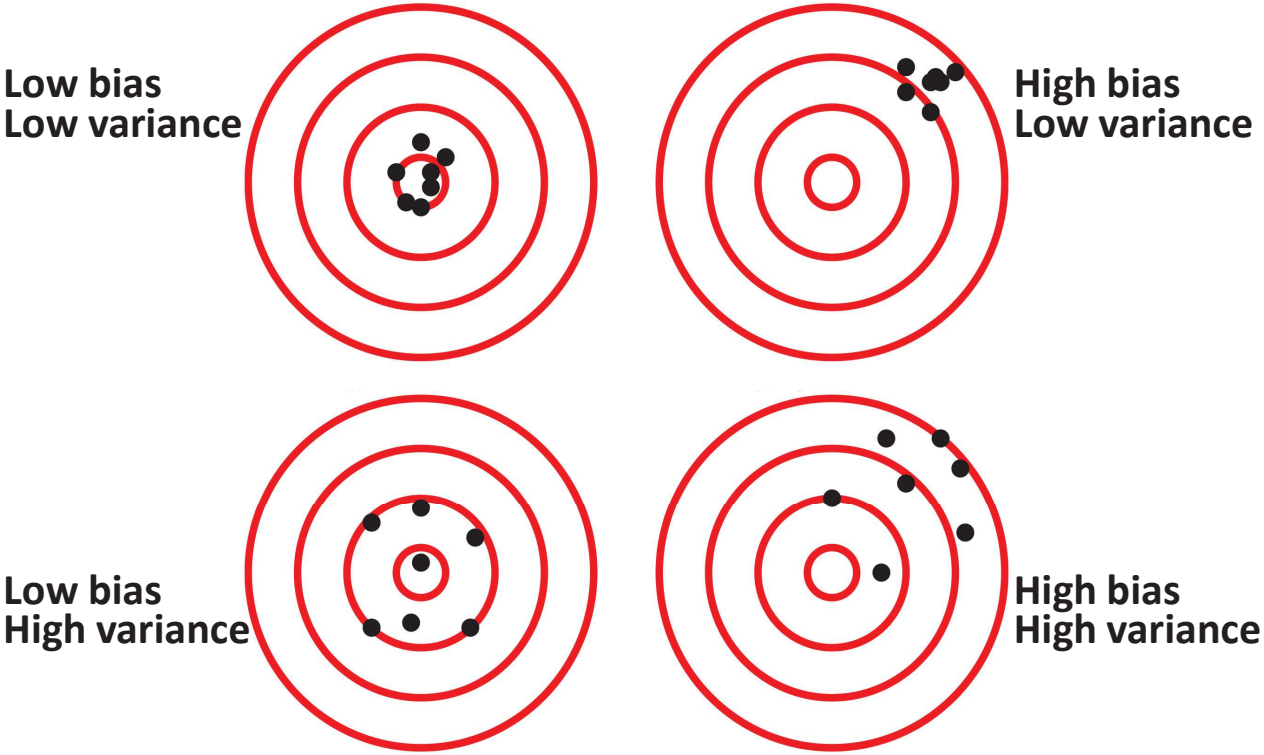
Bias-Variance Trade-off

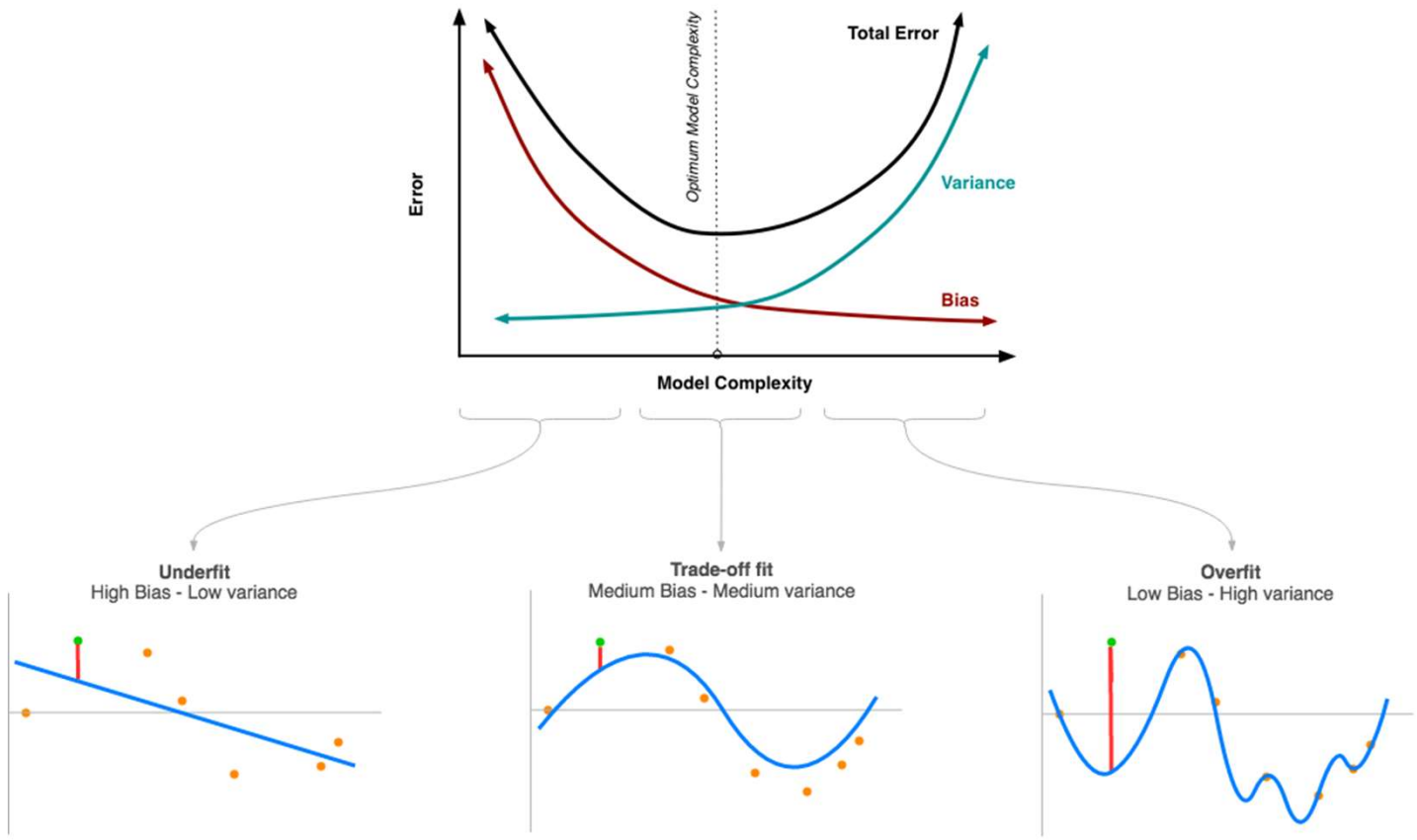
- Models of higher complexity have lower bias but higher variance.
- Once model complexity passes a certain threshold, models “overfit” with the variance term dominating the test error
- From this point onward, increasing model complexity will only decrease performance (i.e., increase test error)
- Once we pass a certain threshold, “larger models are worse.”
- Bias-variance trade off Is revealed via test set not training set

Bias-variance tradeoff using targets

- The goal would be for the points to be near the center of the target.
- To the extent which the points are far from the center, they suffer from bias.
- The variance is the dispersion. The bias and variance combine to form mean squared error



Bias-Variance Trade-off



Source http://www.ebc.cat/author/eduard-bonadagmail-com/)

Double-descent

Deep learning: where the classical bias-variance trade-off breaks down

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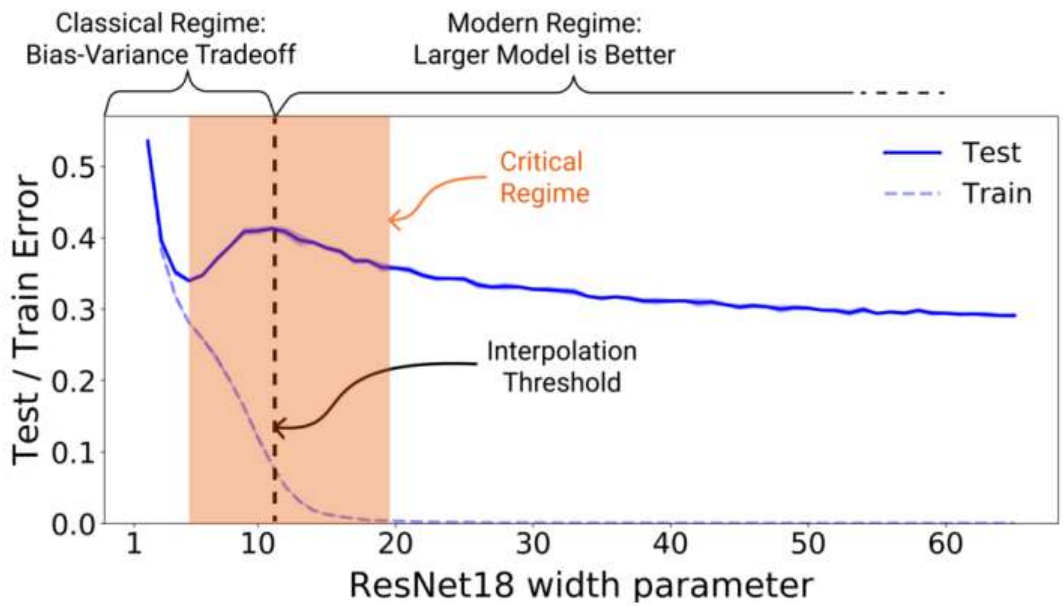
Double-descent

- Observed in neural networks, where the bias-variance tradeoff breaks down
- Test error keeps decreasing as we over-parametrize the network or add more training examples



Double-descent

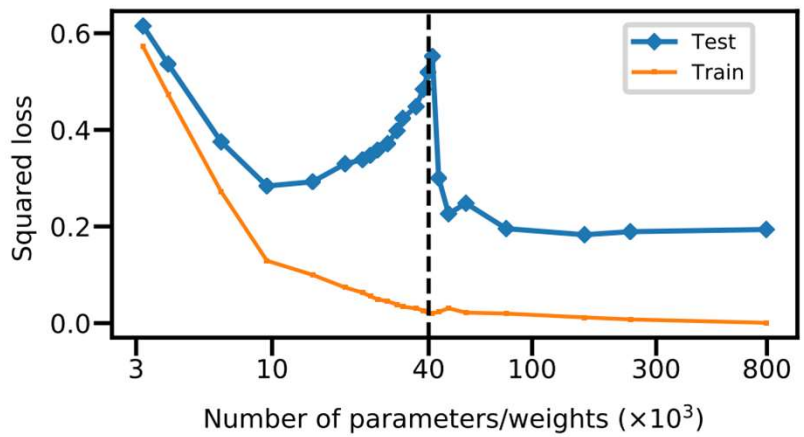


Figure 4: **Double descent risk curve for fully connected neural network on MNIST.** Training and test risks of network with a single layer of H hidden units, learned on a subset of MNIST ($n = 4 \cdot 10^3$, $d = 784$, $K = 10$ classes). The number of parameters is $(d+1) \cdot H + (H+1) \cdot K$. The interpolation threshold (black dotted line) is observed at $n \cdot K$.

Reconciling modern machine learning practice and the bias-variance trade-off, Mikhail Belkin, et al https://arxiv.org/pdf/1812.11118.pdf

Double-descent

- Double descent provides an indication that even though models that pass through every training data point are indeed overfitted
- The structure of the resulting network forces the interpolation to be smooth and results in superior generalization to unseen data
- As long as we keep increasing the model complexity, test error keep decreasing
- After certain complexity, the testing error start to be smaller than the sweet spot that we get within the under-parameterization regime

Automated ML (AutoML)

Does AutoML break TANSAAFL ?

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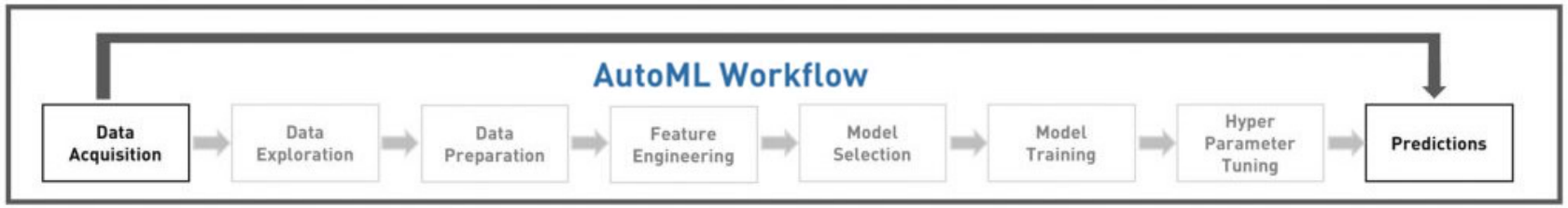
AutoML



Let's consider the no-free lunch theorem again.

If we automate the ML Lifecycle with AutoML, does that mean no-free lunch doesn't apply? (That is can we automate the entire lifecycle of ML & get optimal results always?)

No, no-free lunch theorem still applies because

- AutoML is just optimizing the process – it is not doing creative work
- AutoML will give you faster results – but not necessarily optimal